

Algebra 1
Unit 14
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

High school students in the United States were invited to complete an online survey in 2020. Part of the survey included questions about superhero powers. More than 1,000 students responded to this survey that included a question about a favorite superhero power. Researchers randomly selected 450 of the completed surveys. The two-way table broken down by gender gives partial results. Use this information to complete questions 1-6.

	To Fly	Freeze Time	Invisibility	Super Strength	Telepathy	Total
Females	49	60	48	1	70	228
Males	51	71	27	25	48	222
Total	100	131	75	26	118	450

1. Complete the table.
2. What fraction of male students chose Invisibility as their superpower?

$$\frac{27}{222} = \frac{9}{74}$$

3. Does the fraction in #2 represent a marginal or joint frequency? Explain.

Joint (male and invisibility)

4. What fraction of total students chose Super Strength as their superpower?

$$\frac{26}{450} = \frac{13}{225}$$

5. Does the fraction in #4 represent a marginal or joint frequency? Explain.

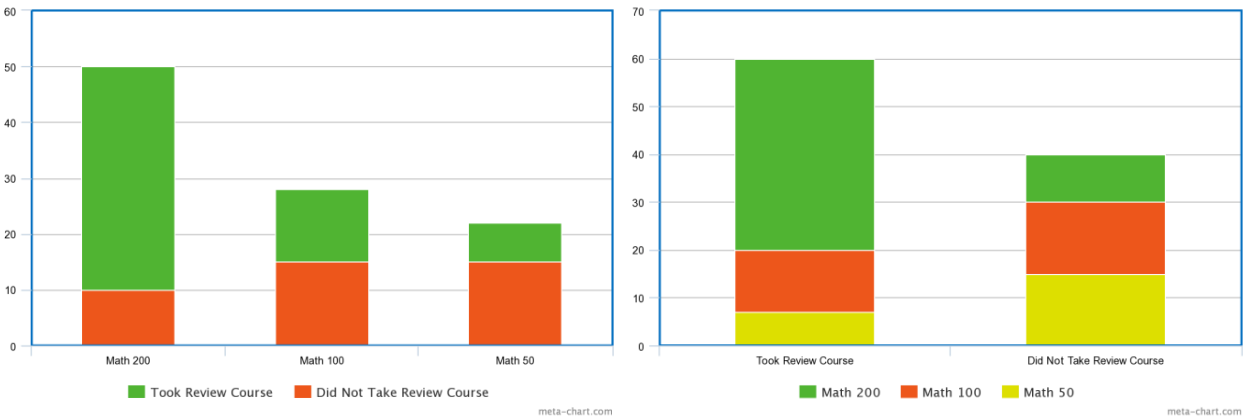
Marginal

6. Is there evidence of an association between gender and a favorite superpower? Explain why or why not.

There are a few superpowers where there is a more significant difference between males and females choosing, so there is an association.

Students were given the opportunity to prepare for a college placement test in mathematics by taking a review course. Not all students took advantage of this opportunity. The following results were obtained from a random sample of students who took the placement test. The results are shown in the two way table and segmented bar graphs.

	Placed in Math 200	Placed in Math 100	Placed in Math 50	Total
Took Review Course	40	13	7	60
Did Not Take Review Course	10	15	15	40
Total	50	28	22	100



7. What fraction of the total surveyed were placed in Math 100?

$$\frac{28}{100} = \frac{7}{25}$$
8. What fraction of students who took the review course were placed in Math 200?

$$\frac{40}{60} = \frac{2}{3}$$
9. Is there evidence of an association between whether a student takes the review course and the math course in which the student was placed? Explain your answer.

Yes, because the majority of students in Math 200 took the review course.
10. Explain which data display was more helpful in determining evidence of an association between students who take the review course and the math course they are placed in.

The last segmented bar because you can more clearly see the increased number of students in Math 200 who took the review course.